

SLECS Explorer User Manual

1. Introduction

SLECS Explorer is an open-source software package for constructing and exploring Skeleton-Level Chemical Space (SLECS) from tandem mass spectrometry (MS/MS) data. The software implements the complete SLECS workflow, including optional spectrum preprocessing, fragment-matrix construction, matrix filtering, non-negative matrix factorization (NMF), SLECS generation, and interactive visualization.

The source code, demonstration datasets, and documentation are publicly available on GitHub, while all datasets and source data associated with this study are available through Zenodo.



2. Running Modes

SLECS Explorer can be used in two ways.

2.1 Online Demo (GitHub Pages)

No installation required.

Suitable for previewing the interface and visualization.

Available at: <https://daffodils520.github.io/SLECS-Explorer/>

The online demo allows users to browse the interface and explore example visualizations. Computational analyses and server-dependent interactive functions are available through the local deployment.

2.2 Local Deployment (Recommended)

Provides the complete functionality of SLECS Explorer.

Clone (or download) the repository and run scripts/app.py.

Recommended for preprocessing, matrix construction, NMF analysis, and fully interactive visualization.

A dedicated online server is planned in future releases to provide the complete interactive functionality through a web browser.

3. Installation

The complete source code is available from the GitHub repository.

Users may either clone the repository using Git or download the complete ZIP package directly from GitHub.

3.1 Obtain the Source Code

Option 1. Clone the repository

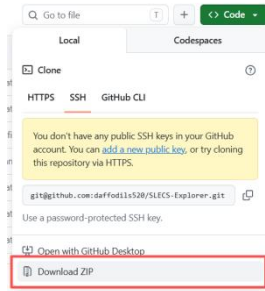
```
<> Bash

git clone https://github.com/daffodils520/SLECS-Explorer.git
cd SLECS-Explorer
```

Option 2. Download ZIP

Click Code → Download ZIP on the GitHub repository page.

After downloading, extract the ZIP package to a local directory.



3.2 First-Time Installation

Open **Anaconda Prompt** and create a dedicated environment:

```
<> Bash

conda create -n slecs_explorer python=3.11 -y
conda activate slecs_explorer
```

Enter the SLECS Explorer project directory:

```
<> Bash

cd /d E:\path\to\SLECS-Explorer
```

Replace the example path with the actual location of the downloaded project folder.

Install the required Python packages:

```
<> Bash

pip install -r requirements.txt
```

Launch SLECS Explorer:

```
<> Bash

python scripts/app.py
```

After the server starts successfully, open the local address shown in the terminal (typically `http://127.0.0.1:5000`) using a web browser.

```
http://127.0.0.1:5000
```

Keep the Anaconda Prompt window open while using SLECS Explorer.

3.3 Subsequent Use

The environment and dependencies only need to be installed once. For subsequent use, open **Anaconda Prompt** and run:

```
</> Bash

conda activate slecs_explorer
cd /d E:\path\to\SLECS-Explorer
python scripts/app.py
```

Then open:

```
http://127.0.0.1:5000
```

4. Input Files

SLECS Explorer accepts the following input files:

File	Description
*.mgf	MS/MS spectra exported from MZmine or other software
*.graphml	Molecular networking results generated by GNPS-FBMN (optional)
*.csv	Preprocessed feature table

Representative example datasets are provided in the **demo/** and **input/** folders.

5. Workflow Overview

The complete workflow consists of the following steps:

- ① Data import
- ② Optional spectrum preprocessing
- ③ Matrix construction and filtering
- ④ NMF decomposition
- ⑤ SLECS generation
- ⑥ Interactive exploration and Export

6. Step-by-Step Tutorial

Step 1. Data Preprocessing (Optional)

Upload the MGF and CSV files, set the preprocessing parameters if required, and click Start Analysis.

Data Preprocessing

Data Preparation

Upload MGF File

Choose File
No file chosen

Accepted format: .mgf

Upload Quantification CSV File

Choose File
No file chosen

Accepted format: .csv

Feature Filtering

Tolerance (Da) Intensity Threshold

Diagnostic Peaks (comma-separated, e.g., 157.1012, 159.1168)

157.1012, 159.1168

Values will be automatically parsed into `features[]`

Neutral-loss Filtering

Neutral-loss exact mass (e.g., H₂O: 18.01528) Neutral-loss type Tolerance (Da)

single loss
v

Intensity Threshold Fragment m/z cutoff (Da)

Start Analysis

Step 2. Matrix Construction

Upload the filtered MGF file, set the matrix construction parameters, and click Start Analysis to generate the fragment-intensity matrix.

Matrix Construction & Optimization

Matrix Construction

Data Preparation (MGF)

Choose File
No file chosen

m/z Binning

Bin size (Da)

0.01

Rule-based Fragment Filtering

Tolerance (Da)

0.02

Elemental Composition: CH or CHO
 Structural Ions: [M+H]⁺, [M+Na]⁺, [M+NH₄]⁺, [M+H-H₂O]⁺, [M+H-2H₂O]⁺
 Structural Mass Differences: Na: 21.981942, NH₃: 17.026547, H₂O: 18.010565

Start Analysis

Step 3. Adaptive Filtering

Upload the intensity matrix, adjust the filtering parameters if necessary, and click Start Analysis to obtain the filtered matrix V (final matrix).

Frequency-based Adaptive Filtering

Upload Step 6 Intensity Matrix (CSV)

[Choose File](#) No file chosen

Description

Use the uploaded CSV to calculate fragment frequencies. The smoothed distribution curve defines adaptive m/z intervals.

- Stricter cutoffs at low m/z remove noise.
- Relaxed thresholds at high m/z preserve rare informative ions.

This balances denoising with signal preservation for structurally relevant features.

[Run](#)

Step 4. NMF Decomposition

Upload the optimized intensity matrix, evaluate the appropriate number of components (k), and perform NMF decomposition to generate the W and H matrices.

NMF Decomposition & Visualization

Evaluation: Select Appropriate components k

Upload Intensity Matrix (CSV)

[Choose File](#) No file chosen

Evaluation Parameters

min_components: 2 max_components: 20 step: 1

[Run Evaluation](#)

NMF Decomposition: Generate matrix W and H

Upload Intensity Matrix (CSV)

[Choose File](#) No file chosen

Same format as evaluation; will decompose by chosen n_components.

Parameters

k_components: 11

[Start Analysis](#)

Step 5. SLECS Visualization

Upload the W matrix (metadata and molecular networking files are optional), adjust the UMAP parameters if needed, and click Start Analysis to visualize the SLECS.

UMAP Visualization

Upload Files

Matrix W (CSV, required) Metadata (CSV, optional) Spectral Similarity Data (.graphml, optional)

[Choose File](#) CNMF_W_matrix_n8.csv [Choose File](#) AS-B2G-0104_quant.csv [Choose File](#) AS-B2G-0104_quant_dedup_filtered

Custom UMAP Parameters

n_neighbors: 10 min_dist: 1.0 random_state: 42

[Start Analysis](#)

Completed

Interactive Visualization: [Open ./umap_visualization/](#)

Step 6. Interactive Exploration and Export

Explore the generated SLECS using the interactive interface. Users can search nodes by Node ID or PEPMASS, select the displayed node information, adjust the cosine similarity threshold interactively, overlay molecular networking, and export figures and analysis results.



7. Notes

Representative example datasets for reproducing the complete SLECS workflow are provided in the input/ directory, while the demo/ directory contains example intensity matrices for the NMF decomposition and visualization modules.

The latest source code, software updates, and documentation are available from the GitHub repository: <https://github.com/daffodils520/SLECS-Explorer>

The complete datasets supporting this study are publicly available through the Zenodo repository: <https://doi.org/10.5281/zenodo.20700190>

8. Contact

For questions regarding the SLECS Explorer software, workflow, datasets, or technical support, please contact:

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9. Citation

If SLECS Explorer is used in published work, please cite the corresponding publication.